

**Conceptual Rover Design for Turkish Lunar Mission****Nurdeniz Altuntaş<sup>a</sup>, Gülcenaz Baygeldi<sup>b</sup>, Beste Boybaş<sup>c</sup>, Furkan Yiğit Demir<sup>d</sup>,  
Barbaros Teoman Kösoğlu<sup>e\*</sup>, Öznur Köşker<sup>f</sup>, Burak Yağlıoğlu<sup>g</sup>**<sup>a</sup> *Geophysical Engineering M.Eng. Student, Istanbul Technical University, Istanbul, Türkiye, [nurdeniz.altuntas@tubitak.gov.tr](mailto:nurdeniz.altuntas@tubitak.gov.tr)*<sup>b</sup> *Aerospace Engineering B.Eng. Student, Middle East Technical University, Ankara, Türkiye, [gulcenaz.baygeldi@tubitak.gov.tr](mailto:gulcenaz.baygeldi@tubitak.gov.tr)*<sup>c</sup> *Physics B.S. Student & Mechanical Engineering B.Eng. Holder, Koç University, Istanbul, Türkiye, [beste.boybasi@tubitak.gov.tr](mailto:beste.boybasi@tubitak.gov.tr)*<sup>d</sup> *Aerospace Engineering B.Eng. Holder, Middle East Technical University, Ankara, Türkiye, [yigit.demir@tubitak.gov.tr](mailto:yigit.demir@tubitak.gov.tr)*<sup>e</sup> *Electrical Engineering B.Eng. Student, Istanbul Technical University, Istanbul, Türkiye, [barbaros.kosoglu@tubitak.gov.tr](mailto:barbaros.kosoglu@tubitak.gov.tr)*<sup>f</sup> *Molecular Biology and Genetics B.S. Holder, Istanbul Technical University, Istanbul, Türkiye, [oznur.kosker@tubitak.gov.tr](mailto:oznur.kosker@tubitak.gov.tr)*<sup>g</sup> *Chief Researcher, TUBITAK UZAY, Ankara, Türkiye, [burak.yaglioglu@tubitak.gov.tr](mailto:burak.yaglioglu@tubitak.gov.tr)*<sup>\*</sup> *Corresponding author*

Moon missions are gaining popularity, with numerous nations, and commercial enterprises aiming to explore the lunar surface. Türkiye is also taking a step to become one of the pioneers in space exploration, with the "National Space Program", introduced in 2021 by the Turkish Space Agency (TUA). One of the main pillars of the National Space Program is the Moon Research Program, where two missions are planned with targets of reaching the Lunar orbit and then the surface through a hard landing within a few years (AYAP-1) and delivering a rover to the surface through soft landing (AYAP-2). In this manner, a group of AYAP-1 scholarship students of TUBITAK UZAY - Space Technologies Research Institute and Space Exploration Society (UKET) members has performed a conceptual design for a small Lunar rover considering AYAP-2 (soft landing mission). The main goal of the designed rover is to carry out scientific research during a lunar day focusing on the physical properties of the landing site, detection and analysis of the concentrations of different minerals and elements, as well as performing a biomining experiment. From the perspective of the biomining experiment, the landing site is considered as the southeast of the Copernicus Crater based on the abundance of iron (Fe), uranium (U), and thorium (Th) elements. A miniature biomining reactor design is studied with an aim of obtaining the maximum elemental mining throughout a lunar day. The investigation of the effects of the lunar conditions on the mining capacity of the microorganisms and performing sustainable space resource mining for in situ resource utilization (ISRU) are considered. Finally, the design factors of the rover platform as well as instruments and payloads are discussed while introducing the technical budgets and concept of operations. Design analyses are introduced via the outcomes of computer simulations which are performed to validate the interaction between the rover and the lunar surface as well as the operational environment.

**1. Introduction**

Türkiye's National Space Program, developed by the Turkish Space Agency (TUA), was announced in February 2020. One of the key component of this program is the Moon Research Program, which will be carried out in two phases. The first phase (AYAP-1) involves lunar orbital operations, the use of a hybrid propulsion system, and a hard landing on the lunar surface [1, 2]. The second phase (AYAP-2) aims to deploy a rover to the Moon via a soft landing [1, 2]. Turkey's goals include enhancing its space technology capabilities, building a deep space heritage for its domestic systems, promoting international cooperation in space science and technology, and raising public awareness of space technologies. TUBITAK UZAY is responsible for designing, developing, and executing the first Turkish Lunar Mission.

This paper proposes a biomining strategy as part of the scientific payload for the Turkish Lunar Rover, aimed at

utilizing elements available on the Moon's surface. While biology and geology experiments have been conducted on previous rovers, space biomining experiments have not yet been performed. Our experimental design is inspired by the European Space Agency's BioRock experiment, conducted on the International Space Station, where bioleaching of rare earth elements from Moon and Mars analog basalts was studied using three bacterial species [3]. Similarly, our design incorporates two bacterial species to investigate their potential for mining uranium and thorium, which hold economic value. To further space biomining research, our experimental design will focus on biomining [4].

This paper also presents a conceptual rover design for the second phase of the Turkish Lunar Mission, planned within the next decade. The rover's subsystems include navigation and mobility, data handling and communication, power, thermal regulation, structure and mecha-

nisms, and the science module. The paper discusses the rover's general features, mission objectives, scientific payloads, and subsystems.

This study builds upon our previous work [4], further exploring the same key principles while expanding the analysis in new directions.

## 2. Design Drivers and Requirements

The design process begins by identifying the mission goals, objectives, and technical requirements. The primary objective is to design a lunar rover capable of traversing the lunar surface and conducting a scientific experiment. The technical requirements are aligned with ECSS systems engineering standards, and the key elements are summarized below:

### 2.1 Functional Requirements

- Perform elemental analysis on the lunar surface using biomining.
- Measure radiation levels on the lunar surface.
- Achieve a speed of approximately 0.12 m/s with 20 cm obstacle clearance.
- Communicate with Earth via the lunar lander, transmitting scientific data.
- Powered by solar panels and onboard batteries.

### 2.2 Mission Requirements

- Deployed at Copernicus Crater.
- Traverse a circular path with a 500 m diameter.
- Stop at three pre-determined sampling points to collect samples.
- Return to its final position within 10 days.

- Perform a bio-rock experiment at the end of the mission.

### 2.3 Environmental Requirements

- Operate at a minimum temperature of 82°C.
- Function between -5°C and 45°C, considering scientific components.
- Withstand uneven lunar terrain and absorb vibrations.

### 2.4 Physical Requirements

- Maximum weight shall not exceed 30 kg.
- The rover shall fit within a 25x45x65 cm volume.
- The rover's solar panel shall be positioned and fixed at an angle to optimize power output.

### 2.5 Configuration Requirements

- The rover shall feature a rocker-bogie suspension.
- The rover shall include a drilling mechanism.
- The rover shall have six wheels.
- The rover shall be equipped with one backup battery.
- The rover shall use six brushless motors.
- The rover shall have an antenna.

After defining these key requirements, the preliminary design was divided into six main subsystems. A product tree of the design, organized by subsystems, is provided in Figure 1.

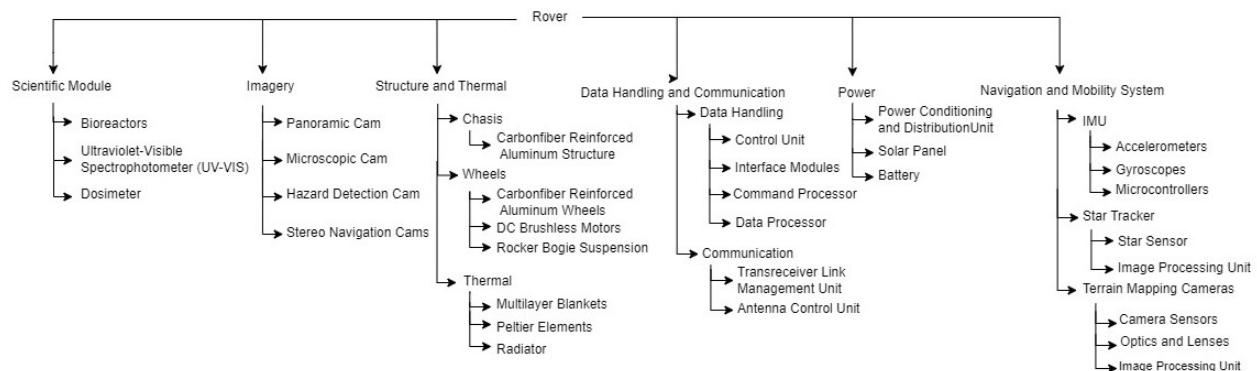


Fig. 1. Product Tree

Table 1. Overview of the key sensors and instruments utilized in lunar exploration missions. The table categorizes these sensors by type, providing details on their specific models and listing the lunar missions in which they were employed.

|                                      | Sensor/Instrument                                     | Utilized By                                                                                 | Reference     |
|--------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------|
| <b>Camera</b>                        | Panoramic Camera (PanCam)                             | Lunokhod 1 and 2, Chang'e 3 (Yutu), Chang'e 4 (Yutu-2)                                      | [5], [6]      |
|                                      | Navigation Cameras (NavCam)                           | Apollo Lunar Roving Vehicle (LRV), NASA's VIPER Rover, Chang'e 3 (Yutu), Chang'e 4 (Yutu-2) | [7], [8], [6] |
|                                      | Hazard Avoidance Cameras (HazCam)                     | NASA's VIPER Rover, Chang'e 3 (Yutu), Chang'e 4 (Yutu-2)                                    | [8], [6]      |
|                                      | Far Ultraviolet Camera/Spectrograph (UVC)             | Apollo 16                                                                                   | [9]           |
| <b>Spectrometers</b>                 | Alpha Particle X-ray Spectrometer (APXS)              | Chang'e 3 (Yutu), Chang'e 4 (Yutu-2)                                                        | [10]          |
|                                      | Visible and Near-Infrared Imaging Spectrometer (VNIS) | Chang'e 3 (Yutu), Chang'e 4 (Yutu-2)                                                        | [10]          |
|                                      | Neutron Spectrometer                                  | Lunar Prospector, NASA's VIPER Rover (planned)                                              | [11], [8]     |
|                                      | Gamma-Ray Spectrometer                                | Lunar Prospector, NASA's VIPER Rover (planned)                                              | [11], [8]     |
|                                      | Alpha Scattering Surface Analyzer (ASSA)              | Apollo 11, 12, 14, 15, 16                                                                   | [9]           |
| <b>Seismometers</b>                  | Lunar Seismic Profiling Experiment (LSPE)             | Apollo 14, 15, 16, 17                                                                       | [12]          |
|                                      | Passive Seismic Experiment (PSE)                      | Apollo 11, 12, 14, 15, 16, 17                                                               | [12]          |
| <b>Particle Detectors</b>            | Alpha Particle Detector (APD)                         | Lunar Prospector                                                                            | [11]          |
|                                      | Ion Mass Spectrometer (IMS)                           | Apollo 12, 15                                                                               | [12]          |
|                                      | Lunar Dust Detector (LDD)                             | Apollo 11                                                                                   | [12]          |
|                                      | Charged Particle Lunar Environment Experiment (CPLLE) | Apollo 14                                                                                   | [12]          |
| <b>Magnetometers</b>                 | Lunar Surface Magnetometer (LSM)                      | Apollo 12, 15, 16, Chang'e 4 (Yutu-2)                                                       | [13], [6]     |
|                                      | Magnetometer                                          | Lunar Prospector, Chang'e 4 (Yutu-2)                                                        | [11], [10]    |
| <b>Thermal and Heat Flow Sensors</b> | Heat Flow Experiment (HFE)                            | Apollo 15, 17                                                                               | [12]          |
|                                      | Thermophysical Properties Probe (TPP)                 | Lunokhod 1, 2                                                                               | [14]          |
| <b>Radiation Detectors</b>           | Radiation Dosimetry Experiment (RDE)                  | Apollo 14, 15, 16, 17                                                                       | [12]          |
|                                      | Cosmic Ray Detector (CRD)                             | Apollo 14, 15, 16, 17                                                                       | [12]          |
| <b>Radar and Lidar</b>               | Lunar Penetrating Radar (LPR)                         | Chang'e 3 (Yutu), Chang'e 4 (Yutu-2)                                                        | [15]          |
|                                      | LIDAR                                                 | NASA's VIPER Rover (planned)                                                                | [8]           |
| <b>Miscellaneous</b>                 | Penetrometer                                          | Apollo 14, Apollo 16, Lunokhod 1                                                            | [12], [5]     |
|                                      | Traverse Gravimeter Experiment (TGE)                  | Apollo 17                                                                                   | [12], [16]    |
|                                      | Diviner Lunar Radiometer                              | Lunar Prospector (Orbiter-USA)                                                              | [11]          |
|                                      | Electron Reflectometer (ER)                           | Lunar Reconnaissance Orbiter (LRO)                                                          | [16]          |
|                                      | Laser Ranging Retroreflector (LRRR)                   | Apollo 11, 14, 15                                                                           | [17]          |
|                                      | Lunar Orbiter Laser Altimeter (LOLA)                  | NASA's VIPER Rover (planned)                                                                | [18]          |
|                                      | Inertial Measurement Units (IMU)                      | Apollo 11, 14, 15                                                                           | [12]          |

## 2.6 Overview of Sensors and Instruments used in Lunar Rovers(Missions)

By outlining each type of sensor and instrument, previously used in lunar missions, mainly in lunar rovers in Table 1, important insights into the rover's design and operation can be obtained. Observing the strengths and weaknesses of sensors and instruments in various environmental settings will help the next steps in the rover design be carried out better and more qualified in comparable missions. This helps rover designers determine which sensors should be a part of the new rover design, to verify good performance and reliability. The type, sensitivity, and range of the sensors and instruments used on the moon have some differences when comparing them. Cameras have been taking high-resolution images of the lunar surface and light interactions are being used to identify what elements the materials consist of. Laser measurements of LIDAR can map the terrain and detect obstacles in its path, and radar can be used to look for subsurface features on the moon's surface. These sensors and instruments can also raise details or trace elements, this way their sensitivity may also play a role. In addition, the environmental tolerance for sensors would also be critical due to aspects that must resist lunar conditions. Sensors and instruments should balance resolution and range a new boundary for Lunar Exploration. By incorporating lessons learned from previous missions on sensor selection, placement, and integration, designers would be able to make further improvements that will increase the capability of the rover in data gathering with a high degree of accuracy and allow the rover to navigate its surroundings with greater precision.

## 3. Landing Site and Mission Description

### 3.1 The Copernicus Crater

Craters located on the Moon's surface hold both geological and scientific significance. In particular, craters formed by large meteor impacts are the most important for understanding the Moon's geological history and dynamics. These craters reveal distinct layers that help scientists interpret the Moon's geological lunar timescale and stratigraphic units with events and their deposits [19]. One of the largest impact craters on the Moon is the Copernicus Crater, located southeast of Oceanus Procellarum (9.62°N, 20.08°W), with a diameter of approximately 93 km. The Copernicus Crater's mineral and elemental compositions, ray system, and central peaks hold significant scientific importance. Notably, its mineral and elemental composition is particularly relevant to the Turkish Lunar Rover mission, where biomining is the primary objective.

For the Turkish Lunar Rover mission, the most suitable area is located in the southeast region of the Copernicus

Crater, due to its high thorium (8.15 ppm) and high uranium concentration (2.1 ppm) as seen in Figure 2 and important physical properties, such as slope.

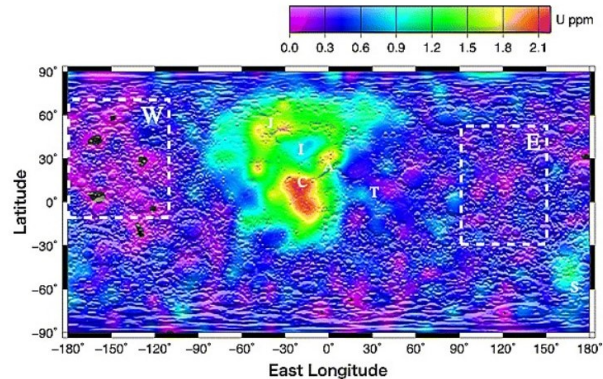


Fig. 2. Uranium Abundance in the Moon. A, the Apenine Bench; C, Copernicus; I, Mare Imbrium; J, Montes Jura; S, South Pole-Aitken Terrane; and T, Mare Tranquillitatis [20]

### 3.2 Description of the Mission

The main goal of this mission is to biomine uranium and thorium on the lunar surface. To achieve this, the rover will be equipped with a sample collection system, a science module, a biosensor device, cameras, sensors, and instruments such as a UV-VIS spectrophotometer and a dosimeter. Data collected from these instruments will be used to analyze the results of the biomining experiments and to create a map and model of the radiation environment on the southeast of the crater. This information will help to understand better the potential opportunities, risks, and challenges of future lunar exploration, resource utilization, and habitation, as well as to develop strategies to protect astronauts and infrastructure.

### 3.3 Trajectory for Lunar Rover

The mission will last for half of a lunar day, equivalent to 10-12 Earth days during the Moon's daytime. A flat area in the southeast region of the Copernicus Crater (Figure 3) has been selected as the soft landing site for the Turkish Lunar Rover mission due to its uranium and thorium abundance and physical properties.

The lunar rover is planned to operate within a 0.5 km radius, collecting samples from various locations within this area. Given the rover's speed limitations, sample collection is expected to be completed within two days. After landing at the central point (7.746°N, 19.240°W), the rover will travel to the designated sampling sites in the following order: Sampling Site-1 (7.739°N, 19.247°W),

Sampling Site-2 (7.755°N, 19.240°W), and Sampling Site-3 (7.739°N, 19.231°W). During these missions, the rover will periodically stop to capture panoramic photographs of the site.

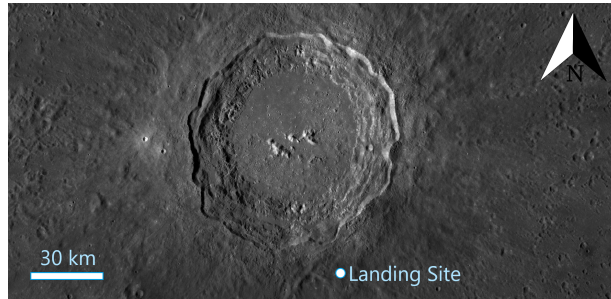


Fig. 3. Landing Site considered as a Turkish Lunar Mission's soft landing phase

## 4. Payloads and Instruments

### 4.1 Dosimeter

Radiation is one of the main considerations during space missions due to its harmful effects on living cells and organisms [21]. Consequently, it is crucial to measure the amount of radiation cells are exposed to in order to assess how it impacts their biomining capabilities and their vulnerability to radiation during the mission. Understanding the radiation levels at the landing site will also provide valuable information for future human space missions. Therefore, it is important to quantify and mitigate exposure to harmful radiation. A dosimeter will be used to measure ionizing radiation at the Copernicus Crater.

### 4.2 Bioreactor

For the investigation of the effects of Moon radiation and low gravity on the fungi performing biomining, miniature bioreactor devices will be utilized. Bioreactors will allow for the generation of a proper setting for selected organisms to grow. They are generally used in biotechnology for the bulk production of industrially important microorganisms. They are adapted for space biomining in order to control and monitor cell cultures [22].

### 4.3 Ultraviolet-Visible Spectrophotometer (UV-VIS)

A UV-VIS spectrophotometer was selected for the mission due to its compact, simple design and its effectiveness in measuring and monitoring microorganism growth via the optical density of fungi cultures. The OD600 measurement will be used to track the turbidity in bioreactors. The basic operation of the UV-VIS spectrophotometer involves the following steps: light is emitted from both a Tungsten lamp, producing visible light, and a D2 lamp,

emitting ultraviolet light. These light beams are combined and reflected from a mirror and directed to a monochromator, which separates the light based on a selected wavelength. This filtered light then passes through both the reference and the sample, and the detector measures the intensity of light absorbed in different wavelengths. By analyzing absorbance variations, the fungi culture concentration will be determined. A lower intensity of light passing through the culture indicates a higher cell density [23].

## 5. Imagery

When designing the camera system for the Turkish Lunar Rover, it is vital to address the mission's scientific objectives and the unique challenges of the lunar environment. To meet these objectives and provide comprehensive coverage of the lunar surface, various types of cameras are strategically positioned across the rover, each serving a distinct purpose.

The Panoramic Camera, located at the front of the rover, captures high-resolution, wide-angle images, offering a comprehensive view of the terrain ahead. This camera is critical for assessing the landscape and planning the rover's movement. The Microscopic Imager, typically positioned lower on the rover's front or side, is designed for close-up analysis, enabling detailed examination of lunar soil, rocks, and potential biological traces. Its function supports key scientific research related to lunar geology and potential life indicators.

To support precise movement and navigation through challenging lunar terrain, Stereo Navigation Cameras are placed at the front and rear of the rover. These cameras generate 3D maps, helping the rover avoid obstacles and make informed decisions during its journey. In addition to these, Scientific Cameras are integrated to assist in various experiments conducted by the rover, providing vital imaging data for different scientific investigations.

In addition to their specialized functions, the cameras must be equipped to handle the harsh lunar conditions, including protection from dust and radiation. Energy efficiency is a crucial consideration, as is data compression to manage the rover's power limitations and storage capacity. By taking these factors into account, the camera system plays a pivotal role in fulfilling the scientific mission and enhancing our understanding of the Moon's geology and environment.

## 6. Science Module

The Moon's surface is an invaluable field for scientific research due to its low gravity and natural exposure to radiation, resulting from the lack of a magnetic field and atmosphere [24, 25]. This makes the Moon an ideal environment to study the effects of low gravity and high radia-

tion on organisms. To date, the major space biominer experiments conducted include Foton M4 - SCK CEN, ESA - BioRock, and BioAsteroid. Biominer involves extracting valuable elements from rocks or regolith by employing microorganisms, which weather the rocks and solubilize cations using organic acids [26, 27]

The BioRock experiment focused on studying bacterial biofilm growth on rocks and microbe-mineral interactions under conditions of ISS. A biominer reactor using bacterial biofilm to extract metals from rocks was deployed. These bioreactors were housed in a centrifuge to simulate Martian gravity and microgravity while growing microorganisms. Upon the experiment's completion, the reactors were returned to Earth, and elemental release rates were analyzed using ICP-MS/OES spectroscopy [28]. The results demonstrated that the bacteria species used significantly increased vanadium leaching across all three gravity conditions [29].

Our conceptual rover mission is designed mainly following the principles of the BioRock experiment. A similar miniature bioreactor will be adapted for in situ sample treatment, with an onboard spectrophotometer conducting the analysis. The rover will be capable of reaching appropriate sampling locations, collecting rock samples, conducting biominer experiments using a fungi species and transmitting the data from cameras, ion detection sensors and a spectrometer. The impact of ionizing radiation and lunar gravity on biominer efficiency and cell viability will be monitored onboard the rover.

Fungi have proven effective in inducing mineral dissolution and mobilizing metals through various mechanisms. The secreted metabolites by fungi and extracellular enzymes interact with substrates. Organic acids, such as citric and oxalic acids, play a significant role in metal solubilization by forming complexes with them [30]. In addition to complexolysis; acidolysis, redoxolysis, and bioaccumulation, also contribute to fungi-based bioleaching, with complexolysis enhancing metal ion solubilization initiated by acidolysis [31].

Our experiment will use the fungal species *Aspergillus niger* to leach Uranium and Thorium. This species is ideal due to its ability to form spores and remain metabolically inactive, faster leaching rate than bacteria, resistance to heavy metals, and production of secondary metabolites that form complexes with metals [32]. *A. niger* is a heterotrophic filamentous fungus with a mycelial structure. Bioleaching fungi are generally mesophilic, performing leaching at 25-35°C. *A. niger* produces remarkable amounts of citric acid and can utilize a wide variety of carbon sources for metabolite production. Alongside carbon, an external nitrogen source is essential for triggering the spore germination [31].

*Aspergillus* is one of the most important and effective genera for extracting valuable metals [33]. Several studies have demonstrated *A. niger*'s ability to leach uranium and thorium. For instance, Amin et al. [34] achieved a uranium recovery rate of 71.4% from uraniferous sandy dolomite, detecting stable organo-uranyl complexes. Similarly, Osman et al. [35] observed a recovery rate of 69% aided by the siderophore produced by *A. niger*. Sun et al. [36] achieved 80% uranium recovery from a Chinese uranium mine. Subsequently, Li et al. [32] enhanced uranium leaching by using glass beads as mechanical stress and a shaker, obtaining a 75.5% recovery rate. Lastly, Wang et al. [37] reached 71.41% uranium recovery and observed structural damage to the ore surface caused by the mechanical force applied by fungal hyphae and organic acids. In addition to uranium, the leaching of thorium by *A. niger* has been investigated in several articles. Keekan and Jalondhara [38] reported the maximum thorium bioleaching rate as 182 µg/L in certain media and culture conditions. Brisson et al. [39] recovered several rare earth elements including Th from monazite sand. The second research investigated the removal of Th ions from radioactive wastewater by using a biosorbent which employs immobilized *A. niger* [40]. Castro et al. [41] used monazite for bioleaching by *A. niger* and proved that fungi could leach and effectively adsorb Thorium. Another study using monazite detected extracted Th in 0.026 and 0.0175 mg/L concentrations [42]. Thus, *Aspergillus niger* species is proven to perform biominer of uranium and thorium.

To initiate the biominer process, the spores of fungi species should be rehydrated. The asexual reproductive structures of fungi are known as conidia or spores [43]. Germination of conidia is required to form germ tubes and hyphae, and eventually to grow and disseminate. Germination occurs in two stages: initial spore swelling by water uptake by C compounds, followed by the outgrowth of hyphae, and the formation of germ tubes by N and C compounds. Certain carbon and nitrogen compounds trigger germination. For example, D-glucose and L-amino acids such as L-serine or L-proline can induce germination within 24 hours [44, 45]. A nitrogen compound will be introduced to break dormancy, allowing control when spores are activated, in this case when the rover completes landing.

*A. niger* cultures will be grown and dormant spore suspension will be obtained for placement in the bioreactors. A convenient dilution of fungi conidia will be reached for all bioreactors to compare growth from the same starting point. The bioreactors will contain three chambers: two for culturing media and dormant conidia, and a third for the rock/regolith sample to be collected. *A. niger* will germinate when the lid between the culture media and fungi

spore chambers is opened. The germination-triggering nitrogen compounds inside the culture media will cause *A. niger* to germinate while the rover performs sample collection. When the samples are delivered to the sample chamber, the activated fungi culture will be in contact with the rock/regolith sample after the second lid opens. The temperature will be maintained between 25-35°C for optimal fungal growth. Each bioreactor will be designed to provide the necessary environmental conditions for *A. niger* species. The biomining process will last for 8-10 Earth days.

To confirm the leaching of ions by *A. niger*, a UV-VIS spectrophotometer and a luminescent Metal-Organic Framework (MOF) located inside the sample chamber will be used to detect the extracted ions. MOFs are a type of complexes having one, two, or three-dimensional structures and are formed by the arrangement of metal ions or metal clusters conjugated together with organic ligands [46]. The adjustable porous property of them allows ions to be captured and interact with the framework, and decrease the luminous intensity as the contacted ions increase, causing fluorescence quenching [47, 48]. A MOF material such as JXNU-4 will contain side groups having a high tendency for U(VI) and Th ions. The mentioned MOF nanomaterial allows analysis in the range of ultraviolet-visible spectra [49]. Thus, mined ions can be adsorbed and detected in an aqueous solution through a designed MOF material.

To compare the effects of gravity on the experiment, the same setup will be replicated on Earth and the ISS. Experiments will be conducted under Earth gravity, Moon gravity, and ISS microgravity conditions with triplicates. Non-biological controls will confirm that the metal dissolution is due to fungi and not abiotic factors.

In summary, our mission will test the ability of *Aspergillus niger* to leach uranium from rock samples collected from the outer walls of Copernicus crater. As a step forward in space biology and resource utilization, this rover mission aims to advance biomining research in space. Further development of miniature bioreactors is essential to ensure proper sealing, pressure retention, and temperature regulation during lunar operations, potentially moving space biomining technology one step closer to feasibility.

## 7. Sample Collecting Mechanism

The lunar surface contains rock fragments, minerals, and elements that will be used for the biomining experiment. To collect the necessary samples, a drilling mechanism was chosen over grasper arms due to its compact design and ease of placement. Unlike grasper arms, drilling mechanisms are shielded from direct sunlight but gener-

ate significant heat during operation, necessitating a heat dissipation system. To maximize the downward force, the drill is placed at the center of mass of the rover. In conclusion, considering the resources on the literature, the core drill mechanism is used in Apollo 11 and Luna 16 missions served as references for this study [50].

In the mission, three sampling locations on the outer walls of the Copernicus Crater were selected, with the on-board drill mechanism employing a flexible tube coring principle. The system consists of a flexible tube, core tube, drill pipe, and sealing device. As the drill rotates and moves downward, the flexible tube encircles the soil core and captures it within the core tube at the desired depth. The sealed tube is then extracted by the transfer mechanism [51], aiming for a depth of approximately 30 cm, as illustrated in Figure 4.

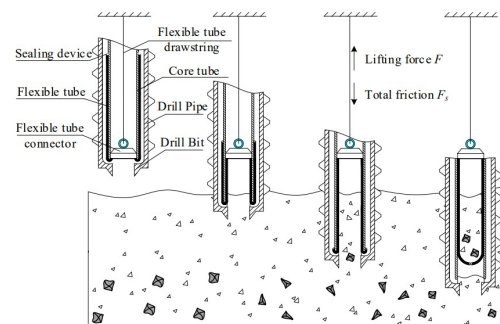


Fig. 4. Flexible Tube Coring Principle [51]

Besides its applicability, the core drill mechanism faces challenges such as friction between components, which creates a negative dragging force, and the nonhomogeneous lunar surface affecting coring quality. To address these issues, a percussive mechanism, similar to the one used in Luna 24, was integrated to increase penetration force [50].

## 8. Navigation and Mobility System

### 8.1 Navigation

The mission's navigation system for the lunar rover is a critical component that ensures accurate movement and positioning on the Moon's surface. This system may include a combination of advanced technologies such as a star tracker, inertial measurement unit (IMU), and terrain mapping cameras. An inertial measurement unit can detect the rover's acceleration and rotation, providing crucial data for precise navigation. A star tracker can be used to locate the rover by comparing the positions of stars in the sky with a known star map, enabling the rover to determine its exact position and orientation relative to the

lunar surface. Additionally, terrain mapping cameras play a pivotal role in assisting the rover's navigation. These cameras capture visual data and create detailed 3D maps of the lunar landscape, helping the rover identify obstacles, plan routes, and ensure safe traversal through challenging terrains.

The inertial measurement unit (IMU) detects the rover's acceleration and rotation, providing data for precise short-term navigation. The star tracker locates the rover by comparing star positions with a known star map, enabling accurate absolute positioning. Meanwhile, terrain mapping cameras capture visual data and create detailed 3D maps of the lunar landscape, assisting in obstacle detection and navigation.

To guarantee the mission's success, a combination of these navigational tools can be utilized. Each system has its own advantages and limitations. The star tracker is highly accurate for absolute positioning, but can face challenges in situations with limited visibility or when the stars are obstructed. The IMU can provide real-time data for short-term navigation, but it may drift over time, leading to positional errors. Visual odometry, based on the data from the terrain mapping cameras, allows the rover to estimate its motion by comparing consecutive images, but it can be affected by visual disturbances like lighting changes and surface texture variations. By integrating these different systems and employing sophisticated algorithms to fuse their data, the rover can overcome individual limitations and achieve robust and reliable navigation on the lunar surface. This comprehensive navigation system ensures that the rover efficiently accomplishes its scientific objectives and successfully explores the Moon.

## 8.2 Wheel Design and Suspension

The designated landing site, Copernicus Crater, is relatively flat, but obstacle clearance for the rover is designed for extreme conditions. To handle the uneven lunar surface, a rocker-bogie suspension system was selected, with two suspensions placed on either side around the drill system. This configuration levels the rover, distributes the load evenly across the 6 wheels, and enhances stability. DC brushless motor is used in each wheel so that each wheel can be steered independently of the other.

The rocker-bogie suspension allows the rover to clear obstacles up to 1.5 times the wheel diameter [52]. To clear 0.2 m obstacles, the minimum wheel diameter must be 0.133 m. Another key factor is drawbar pull—the force available to pull or push an extra weight before reaching maximum traction [53]. This is calculated as the difference between traction force and motion resistance, providing insights into motion characteristics. To calculate the drawbar pull, equation 1 is used with lunar coefficients,

while Martian crater properties are referenced for unavailable data. The results, plotted using four different diameters and five different widths, are shown in Figure 5.

$$z_{rw} = \frac{1}{N} \left( \frac{3W_W \cos \theta}{(3-n)(k_c + b_w k_\varphi) \sqrt{d_w}} \right)^{\frac{2}{(2n+1)}} \quad [1]$$

|                     |                                     |
|---------------------|-------------------------------------|
| cohesive modulus    | $k_c = 1400 \text{ N m}^{-2}$       |
| exponent of sinkage | $n = 0.8$                           |
| wheel diameter      | $n = 0.14 \text{ m}$                |
| cohesion            | $c = 170 \text{ N m}^{-2}$          |
| weight on moon      | $W_W = 48.6 \text{ N}$              |
| frictional modulus  | $k_\varphi = 8200 \text{ N m}^{-2}$ |
| wheel width         | $b_w = 0.08 \text{ m}$              |
| number of wheels    | $N = 6$                             |
| slope angle         | $\theta = 30^\circ$                 |

$$H = c \left( \frac{\pi}{4} \right) \left( \sqrt{z_w d_w} \right) b_w + W_W \cos \theta \quad [2]$$

$$R_r = f_r W_W \quad [3]$$

$$R_c = \frac{1}{n+1} b_w \left( \frac{k_c}{b_w} + k_\varphi \right) z^{n+1} \quad [4]$$

$$R_g = \frac{W_w \sin \theta}{N} \quad [5]$$

$$DP = F - R_{all} = H - R_c - R_g - R_r \quad [6]$$

from these equations it is calculated that;

|                                 |                          |
|---------------------------------|--------------------------|
| soil thrust                     | $H = 40.1 \text{ N}$     |
| soil compaction resistance      | $R_c = 0.0993 \text{ N}$ |
| drawbar pull(neglecting others) | $DP = 33.22 \text{ N}$   |
| rolling resistance              | $R_r = 2.43 \text{ N}$   |
| gravitational resistance        | $R_g = 4.65 \text{ N}$   |

where  $z = 0.0055$  and friction coefficient  $f_r = 0.05$

The derivation of these equations was aided by Rigas [52] and Carrier et al. [54].

According to the Figure 5, drawbar pull does not significantly change with size differences. Based on mass, volume, and durability considerations, a diameter of 0.14-0.16 m and a width of 0.08-0.1 m were determined as the safe margin. Ultimately, a wheel diameter of 0.14 m and a width of 0.08 m were selected.

To improve stability, skid steering, as used in Lunakhod rovers, was chosen [55]. Skid steering allows more reliable and stable rotation by varying the speeds of

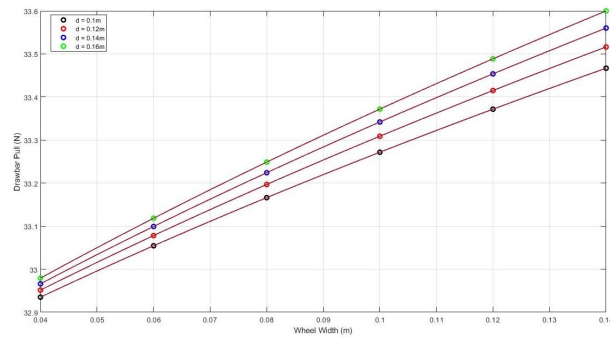


Fig. 5. Drawbar Pull / Wheel Width for 4 different diameters

the wheels. For rotation or turn, the wheels on one side work faster than the other side. This allows the rover to skid on a surface.

## 9. Data Handling and Communication

### 9.1 Data Handling

The Turkish Lunar Rover's data handling system is essential for collecting, processing, and storing data from its scientific instruments, such as images, videos, and measurements.

#### 9.1.1 Data Collection

The rover is equipped with sensors like a UV-VIS spectrophotometer to analyze lunar materials. Cameras, including panoramic, microscopic, and hazard detection cameras, capture visual data from various perspectives.

#### 9.1.2 Data Processing and Storage

Collected data is processed, converting raw information into usable formats. Image processing enhances quality, while reliable high-capacity storage protects the data in the harsh lunar environment.

#### 9.1.3 Data Transmission Speeds

The system ensures efficient transmission of data back to Earth using high-gain antennas and advanced communication protocols when a direct line of sight is available.

### 9.2 Communication

Communication systems are essential for transmitting data from the Turkish Lunar Rover to the ground station through the lander. These systems consist of three key components: the rover, the lander, and the ground station.

- The rover is equipped with a transceiver and antenna system to transmit and receive data, acting as the primary platform for collecting scientific data, images, and telemetry on the lunar surface. It establishes

communication with both the lander and Earth, ensuring the data is sent to mission control.

- The lander serves as an intermediary, housing its own transceiver and antenna to relay data from the rover to Earth. This is necessary because the direct transmission between the rover and the ground station is limited by the distance between the Moon and Earth.
- The ground station on Earth is equipped with a high-gain antenna system, capable of receiving data from the lander, processing it, and sending commands. It plays a crucial role in ensuring the collected data is utilized effectively by the scientific team.

Three communication methods can be considered: radio waves, which are reliable but offer lower data rates; laser communication, providing higher data rates but requiring a clear line of sight and being sensitive to atmospheric conditions; and satellite communication, which allows continuous contact even without a direct line of sight but may introduce higher latency and signal interference.

Carefully choosing the appropriate communication methods ensures the successful transmission of scientific data, supporting the overall mission objectives.

## 10. Structure

In this study, a lunar rover with dimensions of 65x45x30 cm is analyzed. The dimensions are derived based on the size of the solar panels and the wheel/suspension systems. The solar panels measure 65x45 cm, and the wheel diameter is 14 cm. The upper side of the chassis is fully covered by the solar panels to maximize power generation. For the main frame and suspension legs, 6061 or 7075 series aluminum alloys were selected, considering their weight, thermal properties, and strength. Carbon fiber-reinforced plastics (CFRP) are used to reduce weight, particularly for components like wheel rims. It is estimated that the rover will weigh approximately 30 kilograms when fully equipped with payloads. Further iterations aim to reduce the overall weight.

## 11. Power Management

Given the unique location of the Copernicus Crater and the duration of the mission, it is essential to have a reliable and efficient power source that can withstand the lunar environment. The power requirements for the rover have been estimated based on the hourly power usage of its various systems. To ensure a continuous power supply, a solar panel with high-efficiency photovoltaic cells will be used to harness solar energy from the lunar environment. Additionally, a non-rechargeable battery will be integrated into

the power system, it will supply power as a backup to provide enough power to run the rover's systems and instruments for the entire mission duration in case of any issues with the solar panels, or when the generated power is not enough to run all systems. As an alternative to the charged batteries, Radioisotope Thermal Generators (RTGs) can be used to acquire needed power for the rover when the solar panels are defective. RTGs convert the thermal energy released by the radioactive material into electrical energy. Mostly plutonium-238 is used as fuel and in recent years, americium-241 which generates thermal power between 200-240 W is being developed by ESA as an alternative [56]. In space applications and in-situ research missions, RTG is mostly used for thermal management and sleep mode in the lunar night due to vast temperature changes. Developing a heat source consists of multiple layers to ensure safe launch of radioactive material. The system includes fuel, Pt-Rh cladding housing, and carbon composite aeroshell. Since the design and the application require detailed research and development process, it has been decided to use a battery design that has taken place in the literature instead of RTG.

### 11.1 Power Budget

To have a comprehensive understanding of the power requirements for different rover systems, the power consumption budget in full operation mode, scientific mode, and idle mode is provided in Table 2.

| Subsystem               | Full Mode    | Scientific Mode | Idle Mode    |
|-------------------------|--------------|-----------------|--------------|
| Navigation and Mobility | 12 Wh        | 0 Wh            | 0 Wh         |
| Thermal System          | 2 Wh         | 2 Wh            | 2 Wh         |
| Power Management        | 4 Wh         | 4 Wh            | 4 Wh         |
| Communications          | 5 Wh         | 5 Wh            | 2 Wh         |
| Processing              | 5 Wh         | 5 Wh            | 2 Wh         |
| Imagery                 | 9 Wh         | 9 Wh            | 4 Wh         |
| Navigation System       | 8 Wh         | 0 Wh            | 0 Wh         |
| Science Module          | 2 Wh         | 2 Wh            | 0 Wh         |
| <b>Total</b>            | <b>47 Wh</b> | <b>27 Wh</b>    | <b>14 Wh</b> |

Table 2. Power budget

### 11.2 Solar Irradiance and Power Generation

According to Kaczmarzyk et al. [57] the average solar irradiance on the near side of the Moon, where the Copernicus Crater is located, is approximately 1356 W/m<sup>2</sup>. As-

suming a solar panel efficiency of 20% at the given temperatures and using solar panel sizes of 40cm × 60cm = (0.24m<sup>2</sup>), the estimated power output is calculated as follows:

$$P_{out} = \text{Eff} \times \text{Irradiance} \times \text{Area} \quad [7]$$

$$P_{out} = 0.2 \times 1356 \text{ W/m}^2 \times 0.24 \text{ m}^2 = 65 \text{ W} \quad [8]$$

It's important to consider that this solar irradiance value was average, and therefore the actual power produced could vary within a certain range. The final power output is estimated to be 65 ± 4% W, which is greater than our estimated required power.

### 11.3 Power Management and Efficiency Strategies

To ensure efficient power management during the mission, the rover's control software will be designed to minimize energy consumption wherever possible. The power management system will continuously track energy levels, system demands, and operational modes to optimize power distribution. Key power-saving strategies that will be employed include:

- **Low-Power Mode:** The rover will switch to a low-power mode during inactive periods, significantly reducing energy usage.
- **Adaptive Power Distribution:** Power will be allocated dynamically across various subsystems, adjusting in real-time to meet their specific needs.
- **Selective Sensor Activation:** Sensors will only be powered on when required, conserving energy and prolonging their lifespan.
- **Efficient Communication Protocols:** Streamlined communication protocols will be used to reduce the power required for data transmission.

By incorporating these power management techniques, the mission's energy needs can be met, ensuring the rover operates consistently and reliably during its exploration of the Copernicus Crater.

## 12. Thermal Management

Drastic temperature changes are observed on the lunar surface due to the temperature difference between the day and night which reaches almost 250°C. To overcome this issue, a proper mission and thermal design should be implemented. In this study, the mission plan is designed for around a half lunar day (10-12 days). The rover reaches its final destination within 10 days to complete the necessary experiments. For this reason, the thermal design

is focused on adapting warm environment rather than Lunar night conditions to provide an internal temperature balance for electronic devices and biological creatures.

At 45° latitude on the moon, the temperature rises from -87.9 °C at 0.5 days after sunrise, up to 81.5°C during mid-day [58]. In this study, the landing site on the Copernicus Crater is located on the 7.7 °N meridian. Due to the low axial tilt of the moon and the flatness of the landing point, it is predicted that similar thermal conditions are observed in this location.

Certain mission scenarios are analyzed by considering both external and internal factors. External factors include solar radiation, surface albedo, infrared radiation, and heat loss to outer space. Internal factors involve thermal conduction and heat dissipation within the system.

Since the chassis involves most of the electronic devices and the biological experiment, the temperature margin is determined in Table 3, the thermal range for the chassis is determined as -5, +45°C. The design should be further improved by considering the biological creature's requirements.

| Equipment               | Min (°C) | Max (°C) |
|-------------------------|----------|----------|
| Motor                   | -40      | 100      |
| Motor Controller        | -20      | 70       |
| Cameras                 | -40      | 85       |
| CPU                     | -20      | 125      |
| IMU                     | -40      | 100      |
| Battery in Discharge    | -20      | 60       |
| Radio                   | -40      | 85       |
| Power Distribution Unit | -40      | 85       |
| Ethernet Switch         | -40      | 60       |

Table 3. CubeSat Equipment Temperatures

Thermal management has an important role in the efficiency and the lifetime of the components. For this reason, several methods are suggested for thermal management. The first method is to integrate composite materials have less conductivity such as carbon – carbon composites for the places don't require heat dissipation. This method also provides a great reduction in mass while conducting less heat. Heat sinks, heat pipes are considered for heat dissipation systems. Peltier elements can be used to provide both dissipation and insulation. Finally, multi-layer insulation (MLI) should be utilized to protect the rover from the Sun's direct and reflected radiation.

### 13. Conclusion

In this paper, a comprehensive conceptual design study was presented for a potential future mission aimed at exploring the surface of the Moon. The study outlines the proposed rover mission, detailing the scientific experiments to be conducted, alongside a thorough analysis of the landing site and trajectory design. The insights and technical discussions provided are intended to serve as valuable guidance for the upcoming Turkish Lunar soft landing mission, offering a foundation for further development and refinement of mission strategies.

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